

Cryptography

Assignment - 4 (Theory)

1. Cipher text (Decryption)

→ H L D T a F H J D W X Q L H A G

key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ Find Plaintext

Key Determined

$$= 21 - 10$$

$$= 11$$

Inverse of 11.

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$26 = 11(2) + 4$$

$$11 = \cancel{4}(2) + 3$$

$$4 = 3(1) = 1$$

$$1 = 4 - 3$$

$$1 = 4(1) - 4(2)$$

$$1 = 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2) - 11)$$

$$1 = 3(26) - 11(6) - 11$$

$$= 1911$$

H4 P1 Q1 HJ 10x 1w2 1cm 1A6

$$i) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 21 + (14 \times 11) \\ 63 + 55 \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$ii) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 9 + 14(19) \\ 27 + 5(19) \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix}$$

$$iii) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 48 + 112 \\ 144 + 40 \end{bmatrix} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$iv) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 21 + 126 \\ 63 + 45 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix}$$

$$v) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix}$$

$$= \begin{bmatrix} 9 + 14(23) \\ 27 + 115 \end{bmatrix} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix}$$

$$vi) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 10 \end{bmatrix}$$

$$= \begin{bmatrix} 68 + (14 \times 16) \\ 198 + \cancel{180}^{80} \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 14 \\ 18 \end{bmatrix}$$

$$vii) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 3(2) + 14(12) \\ 9(2) + 5(12) \end{bmatrix}$$

$$\begin{bmatrix} 174 \\ 78 \end{bmatrix} \pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix}$$

$$viii) \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 + 84 \\ 0 + 30 \end{bmatrix}$$

$$= \begin{bmatrix} 84 \\ 30 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Plaintext

↳ TOP SECRET MESSAGE

2) consider the Hill cipher key matrix

$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$ cipher text : R X A O D Y
to find Plain text.

Decryption

$$P = (K^{-1} \times C) \bmod 26$$

$$\text{Det}(K) = (24 \times 14) - (19 \times 5)$$

$$= 336 - 95$$

$$= 241 = 7 //$$

7 is coprime of 26.

$$7^{-1} = 15 //$$

$$K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 105 \\ 315 & 360 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$R_x = \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 57 \\ 557 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} \Rightarrow \begin{matrix} F \\ L \end{matrix}$$

$$A_o = \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$D_y = \begin{bmatrix} 3 \\ 24 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

Plaintext = FLOWER.

4) Transposition $\rightarrow$ encrypt

"The arrow is pointing North.
direction" using Simple key = 4

Keyword.

$\rightarrow$ ATLAS.

double

$\rightarrow$ Key 1: BREAK

$\rightarrow$ Key 2: log.

Simple

THEA
ROW
ISPO
INTI
NUNO
RTHD
IREC
TION

Keyword:

ATLAS
15324

THEAR
ROWIS
POINT
INUNO
RTHDI
RECTI
ONXXX

~~Cipher:~~

~~TRPIRO
HOONTEN
EWIGHCX
AINNDIX
RSTOIX~~

Cipher:

TRIINRITHRSN
TRIEOPTNHEO
AUNIDUN

Cipher text

TRPIRROAINNDTXEWLGHCX

RSTOII XHOONTEN

Double.

BREAK

25314

TEAR

ROWIS

POINT

INGNO

RTHDI

RECTI

ONXXX

Cipher.

AINNNTXTRPIRRO

EWLGHCX RSTOII XH

OONTEN

Log.

Log

231

AINN

NDT

TRP

IRR

DEW

LGHCX

CRS

TOI

IHO

ONT

ENX

Cipher

NTPRWHSIOX

ANTIOCTIOEID

RREGROHNNX

keyword $\rightarrow$ SECRET

total = 36

LL ED NH PTR SOIT

key length - 6

MAXSA XERXT

5 2 1 4 3 6
SECRET

ANONI SNFOTX

T H I S I S

A P L A I N

N T E X T F

O P D E N O

N S T R A T

1 0 W X X X

THIS IS A PLAIN

TEXT FOR DEMON

- STATION.

9) Permutation.

Find the Inverse Permutation.

B) 43152

$P = (4, 3, 1, 5, 2)$

1 is in position 3

$2 \rightarrow 5$

$3 \rightarrow 2$

$4 \rightarrow 1$

$5 \rightarrow 4$

Inverse Permutation (3, 5, 2, 1, 4)

ii) 7 4 1 3 5 2 6

$$P = (7, 4, 1, 3, 5, 2, 6)$$

$$1 \rightarrow 3$$

$$5 \rightarrow 5$$

$$2 \rightarrow 4$$

$$6 \rightarrow 7$$

$$3 \rightarrow 1$$

$$7 \rightarrow 1$$

$$4 \rightarrow 2$$

$$P^{-1} = (3, 4, 1, 2, 5, 7, 1)$$

iii) 4 1 3 2

$$P = (4, 1, 3, 2)$$

$$1 \rightarrow 2$$

$$2 \rightarrow 4$$

$$3 \rightarrow 3$$

$$4 \rightarrow 1$$

$$P^{-1} = (2, 4, 3, 1)$$

iv) 3 5 1 4 2

$$P = (3, 5, 1, 4, 2)$$

$$1 \rightarrow 3 \quad 4 \rightarrow 4$$

$$2 \rightarrow 5 \quad 5 \rightarrow 2$$

$$3 \rightarrow 1$$

$$P^{-1} = (3, 5, 1, 4, 2)$$

3. Confusion vs. Diffusion

Analyse the following classical ciphers with respect to confusion and diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext.

Cipher	Does it change the character/symbol value?	Does it change the character position	Confusion / Diffusion	Justification
Shift cipher	Yes	No	Confusion	Substituted via mathematical shift $(p+k)$. Character order is unaltered
Multiplicative Cipher	Yes	No	Confusion	Substituted via modular multiplication $(p \times k)$. Character order is unaltered
Affine Cipher	No	Yes	Diffusion	Characters retain the original value, but their positions are scrambled across the text.
Transposition Cipher	Yes	No	Confusion	Multiplicative & shift structures $((a-p+b) \bmod 26)$ position remains unchanged

Relationship between substitution and confusion:

In a substitution cipher, the position remains the same, but the actual character value changes

$$A \longrightarrow D$$

Therefore, the relationship between the plaintext characters and ciphertext character is hidden. Hence substitution mainly provides confusion.

Relationship between Transposition and Diffusion:

In transposition, the character itself doesn't change.

ABCDE —> CADBE

Therefore, the characters are moved to different positions. This spreads the information of the plaintext throughout the ciphertext. Hence transposition mainly provides diffusion.